

MIC : Masked Image Consistency for Context-Enhanced Domain Adaptation

CVPR, 2023

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01

Introduction

01. Introduction

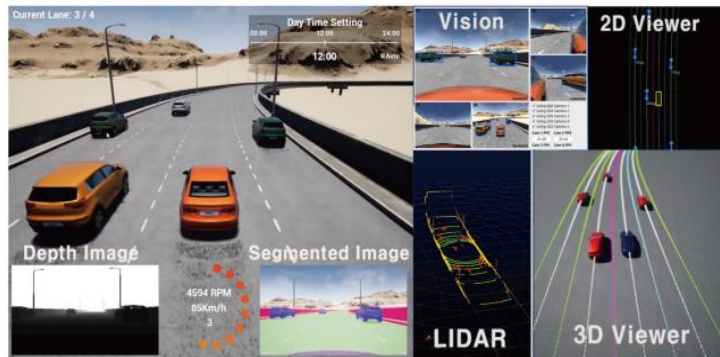
1. Domain Adaptation



01. Introduction

1. Domain Adaptation

▶ 시뮬레이터를 통한 데이터 확보



01. Introduction

1. Domain Adaptation



GTA 5 screenshots



Cityscapes Dataset

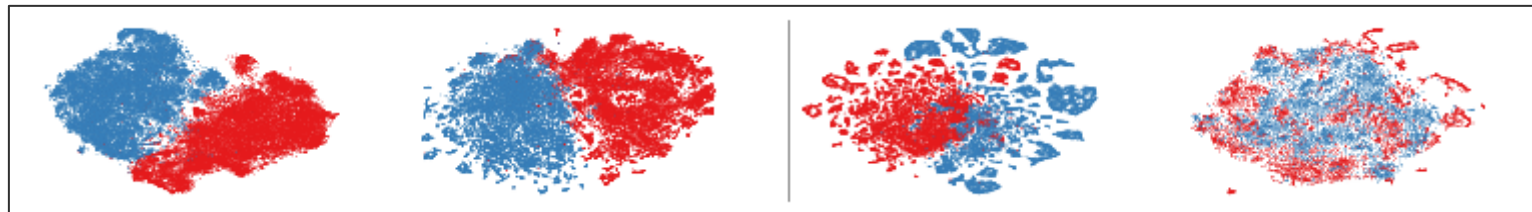


01. Introduction

1. Domain Adaptation

● Cityscape

● GTA5



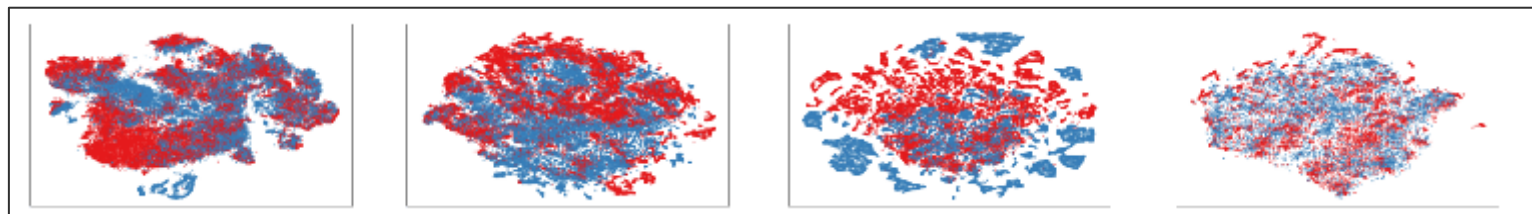
Road

Sidewalk

Wall

Person

Domain Adaptation



Road

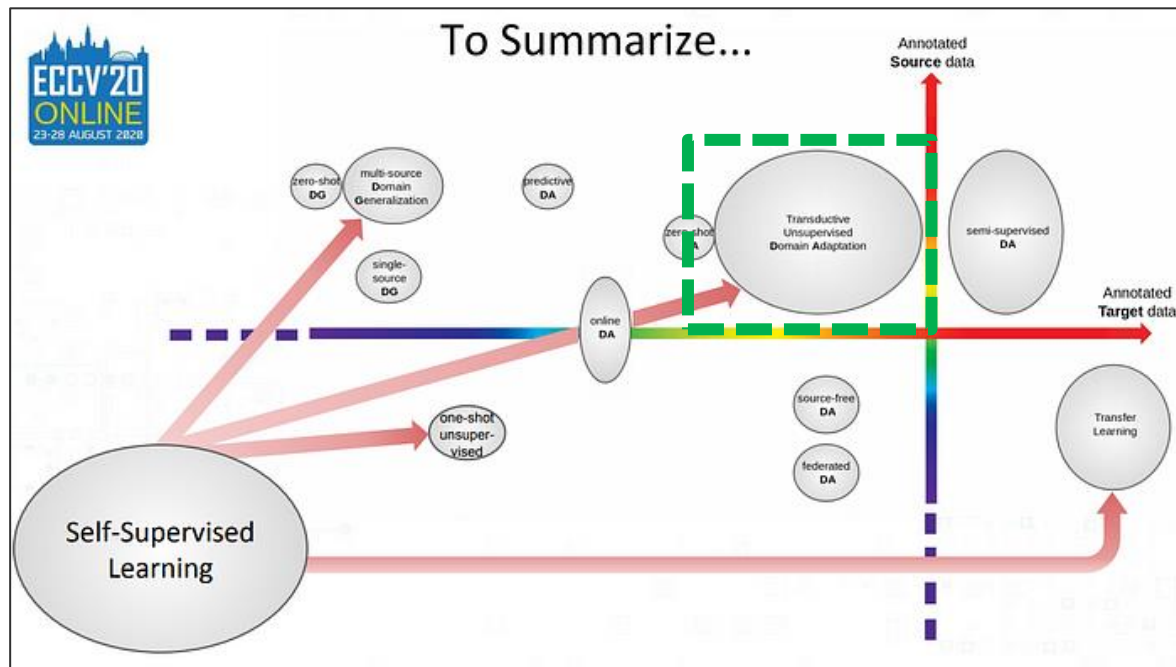
Sidewalk

Wall

Person

01. Introduction

2. Unsupervised Domain Adaptation(UDA)



01. Introduction

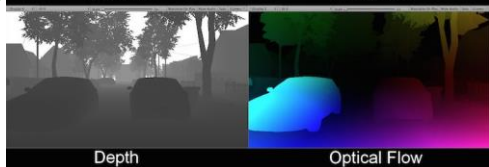
2. Unsupervised Domain Adaptation(UDA)

► Reduce Cost of Labeling

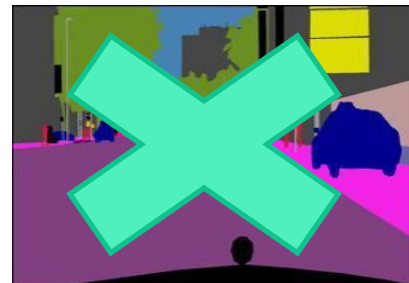
Source : Simulation



FREE



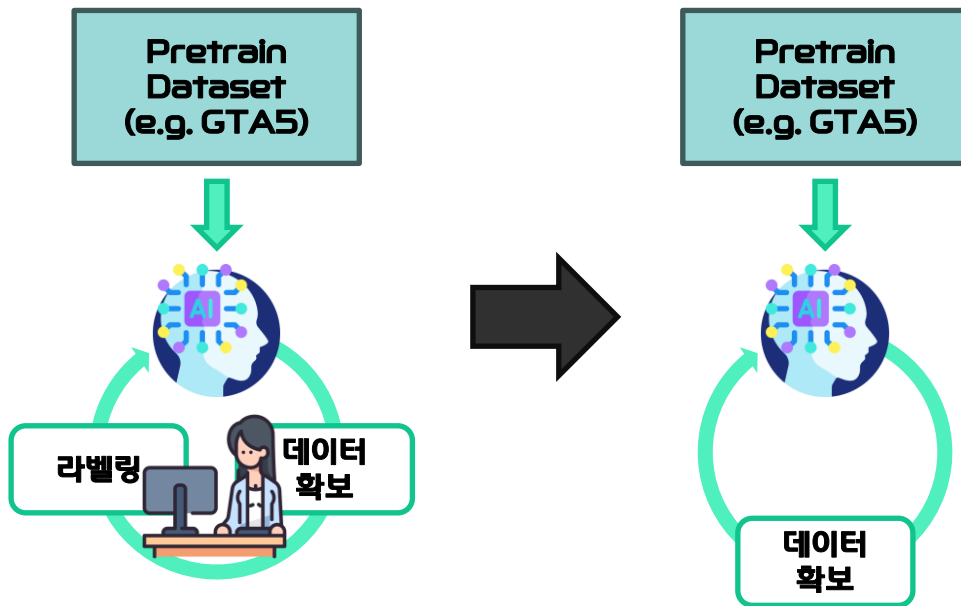
Target : Real



01. Introduction

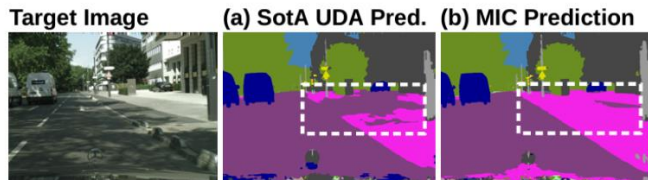
2. Unsupervised Domain Adaptation(UDA)

▶ Self Training



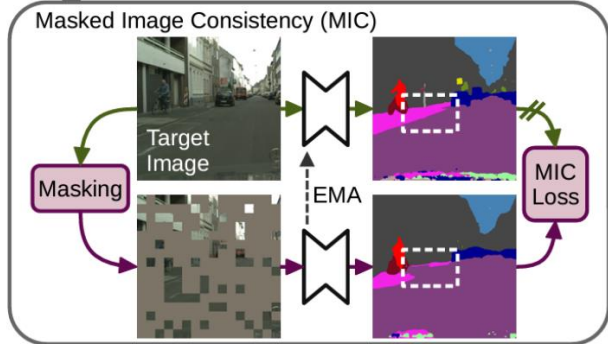
01. Introduction

3. MIC (Masked Image Consistency) Method

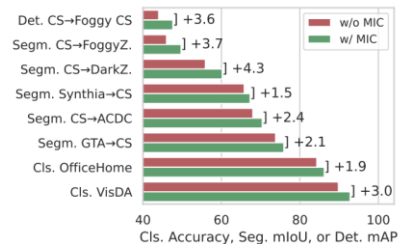


(c) UDA with Masked Image Consistency (MIC)

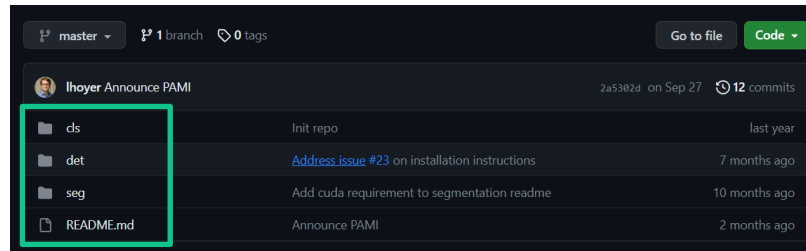
Unsupervised Domain Adaptation (UDA) Method



► Apply to Any UDA Method



► Computer Vision Task



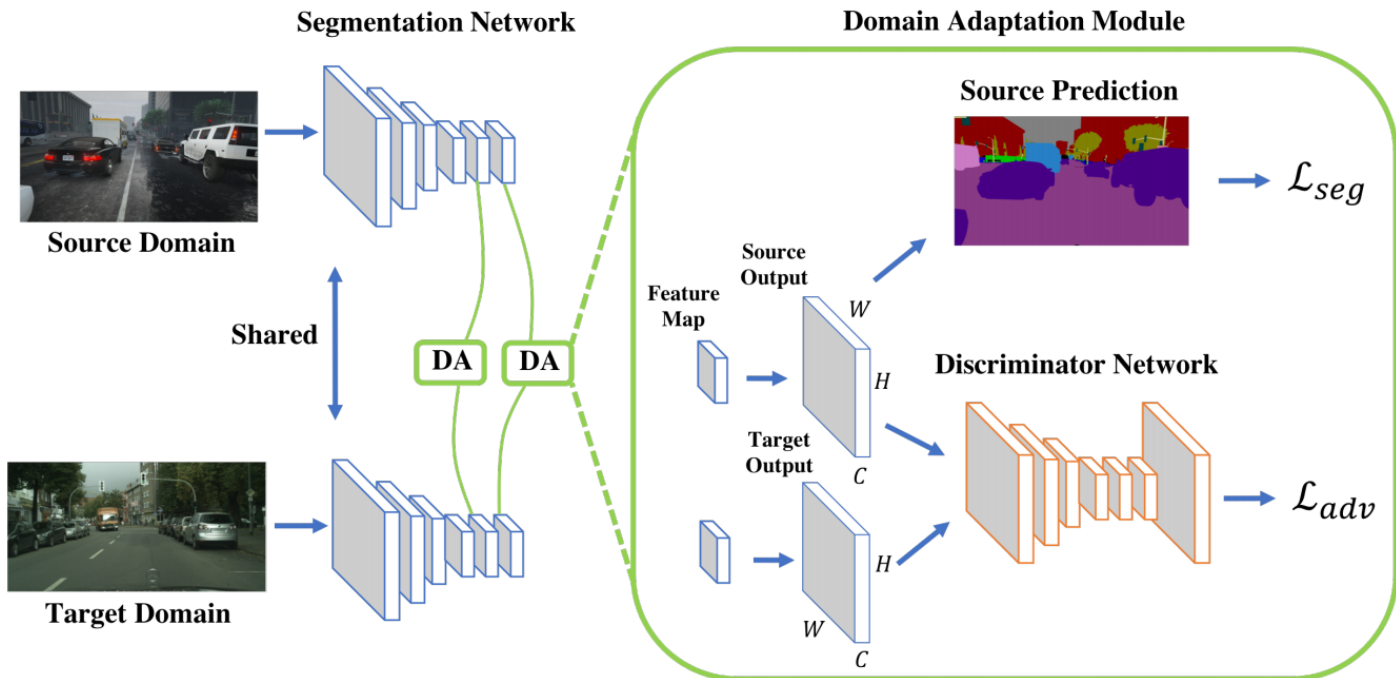
02

Related Works

02. Related Works

1. Learning to adapt structured output space for semantic segmentation, 2018

► Adversarial Method



02. Related Works

2. DAFormer: Improving Network Architectures and Training Strategies for Domain-Adaptive Semantic Segmentation, 2022

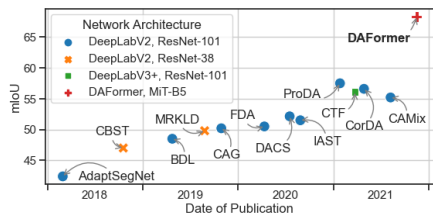


Table 6. Comparison with state-of-the-art methods for UDA. The result

Architecture	Src-Only	UDA	Oracle	Rel.
DeepLabV2 [6]	34.3 ±2.2	54.2 ±1.7	72.1 ±0.5	75.2%
DA Net [18]	30.9 ±2.1	53.7 ±0.2	72.6 ±0.2	74.0%
ISA Net [35]	32.3 ±2.1	53.3 ±0.4	72.0 ±0.5	74.0%
DeepLabV3+ [7]	31.0 ±1.4	53.7 ±1.0	75.6 ±0.9	71.0%
SegFormer [87]	45.6 ±0.6	58.2 ±0.9	76.4 ±0.2	76.2%

	Road	S.walk	Build.	Wall	Fence	Pole	Tr.Light	Sign	Veget.	Terrain	S
GTA5 → Cityscapes											
CBST [100]	91.8	53.5	80.5	32.7	21.0	34.0	28.9	20.4	83.9	34.2	8
DACS [71]	89.9	39.7	87.9	30.7	39.5	38.5	46.4	52.8	88.0	44.0	88.8
CorDA [79]	94.7	63.1	87.6	30.7	40.6	40.2	47.8	51.6	87.6	47.0	89.7
ProDA [94]	87.8	56.0	79.7	46.3	44.8	45.6	53.5	53.5	88.6	45.2	82.1
DAFormer	95.7	70.2	89.4	53.5	48.1	49.6	55.8	59.4	89.9	47.9	92.5

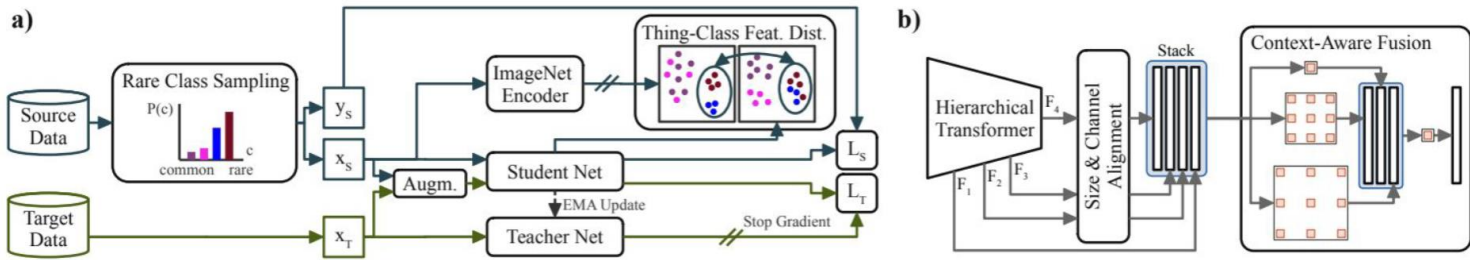


Figure 2. Overview of our UDA framework with Rare Class Sampling, Thing-Class Feature Distance, and DAFormer network.

02. Related Works

3. HRDA: Context-Aware High-Resolution Domain-Adaptive Semantic Segmentation, 2022

(b) Our HRDA Semantic Segmentation Training

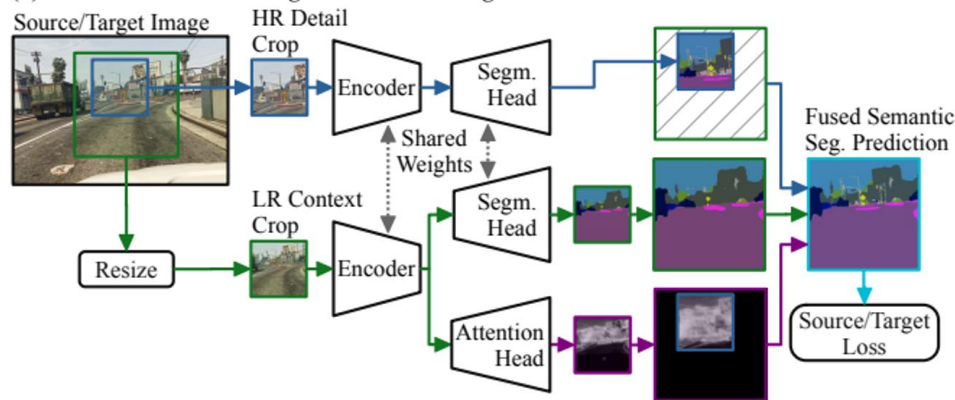


Table 1. Comparison with previous UDA methods. The results of HRDA are averaged over 3 random seeds. Further methods are shown in the supplement.

	Road S.walk Build. Wall Fence Pole Tr.Light Sign Veget. Terrain Sky Person Rider Car Truck Bus Train M.bike Bike mIoU																			
GTA5 → Cityscapes																				
CBST [90]	91.8	53.5	80.5	32.7	21.0	34.0	28.9	20.4	83.9	34.2	80.9	53.1	24.0	82.7	30.3	35.9	16.0	25.9	42.8	45.8
DACS [63]	89.9	39.7	87.9	30.7	39.5	38.5	46.4	52.8	88.0	44.0	88.8	67.2	35.8	84.5	45.7	50.2	0.0	27.3	34.0	52.1
CorDA [71]	94.7	63.1	87.6	30.7	40.6	40.2	47.8	51.6	87.6	47.0	89.7	66.7	35.9	90.2	48.9	57.5	0.0	39.8	56.0	66.6
BAPA [41]	94.4	61.0	88.0	26.8	39.9	38.3	46.1	55.3	87.8	46.1	89.4	68.8	40.0	90.2	60.4	59.0	0.0	45.1	54.2	57.4
ProDA [87]	87.8	56.0	79.7	46.3	44.8	45.6	53.5	53.5	88.6	45.2	82.1	70.7	39.2	88.8	45.5	59.4	1.0	48.9	56.4	57.5
DAFormer [29]	95.7	70.2	89.4	53.5	48.1	49.6	55.8	59.4	89.9	47.9	92.5	72.2	44.7	92.3	74.5	78.2	65.1	55.9	61.8	68.3
HRDA (Ours)	96.4	74.4	91.0	61.6	51.5	57.1	63.9	69.3	91.3	48.4	94.2	79.0	52.9	93.9	84.1	85.7	75.9	63.9	67.5	73.6
Synthia → Cityscapes																				
CBST [90]	68.0	29.9	76.3	10.8	1.4	33.9	22.8	29.5	77.6	-	78.3	60.6	28.3	81.6	-	23.5	-	18.8	39.8	42.6
DACS [63]	80.6	25.1	81.9	21.5	2.9	37.2	22.7	24.0	83.7	-	90.8	67.6	38.3	82.9	-	38.9	-	28.5	47.6	48.3
BAPA [41]	91.7	53.8	83.9	22.4	0.8	34.9	30.5	42.8	86.6	-	88.2	66.0	34.1	86.6	-	51.3	-	29.4	50.5	53.3
CorDA [71]	93.3	61.6	85.3	19.6	5.1	37.8	36.6	42.8	84.9	-	90.4	69.7	41.8	85.6	-	38.4	-	32.6	53.9	55.0
ProDA [87]	87.8	45.7	84.6	37.1	0.6	44.0	54.6	37.0	88.1	-	84.4	74.2	24.3	88.2	-	51.1	-	40.5	45.6	55.5
DAFormer [29]	84.5	40.7	88.4	41.5	6.5	50.0	55.0	54.6	86.0	-	89.8	73.2	48.2	87.2	-	53.2	-	53.9	61.7	60.9
HRDA (Ours)	85.2	47.7	88.8	49.5	4.8	57.2	65.7	60.9	85.3	-	92.9	79.4	52.8	89.0	-	64.7	-	63.9	64.9	65.8

Q & A

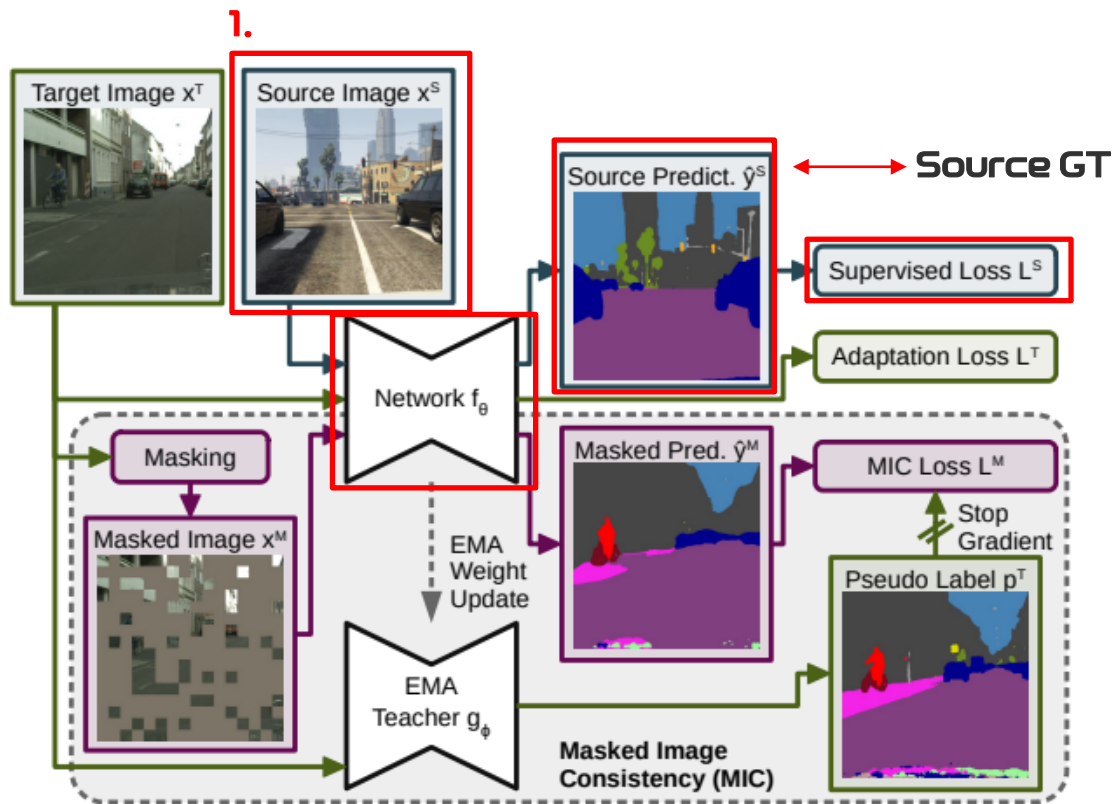
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03

Methods

03. Methods

1. Source Image Process



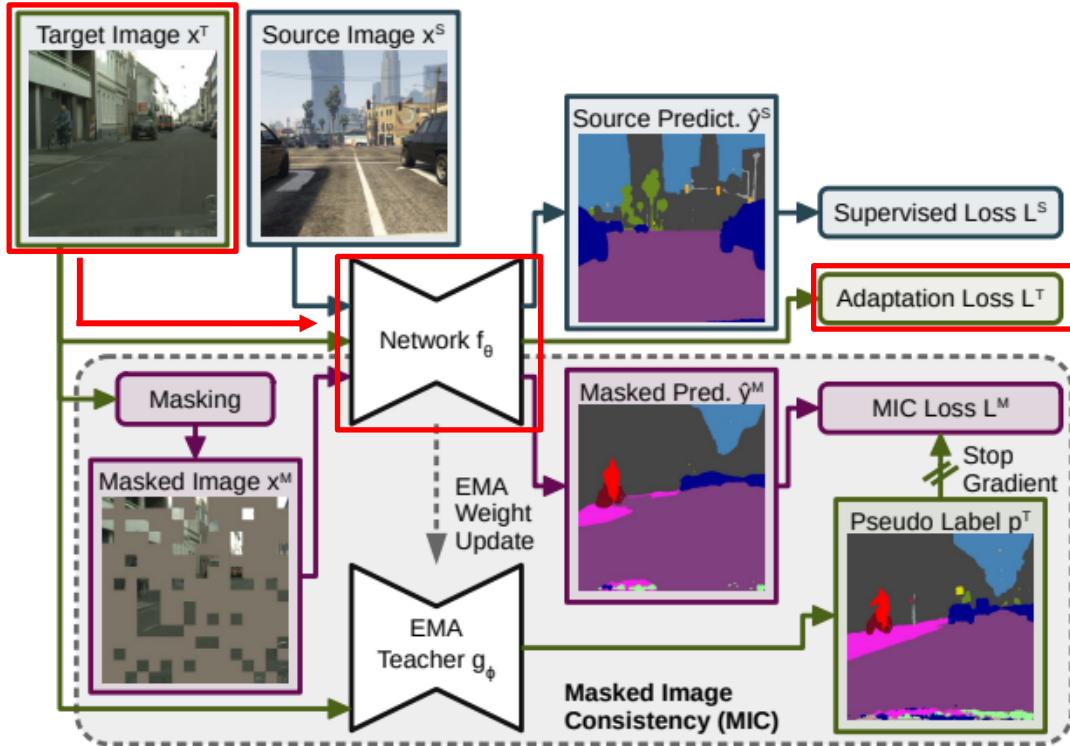
$$\mathcal{L}_k^{S,cls/seg} = \mathcal{H}(f_\theta(x_k^S), y_k^S),$$

$$\mathcal{H}(\hat{y}, y) = - \sum_{i=1}^H \sum_{j=1}^W \sum_{c=1}^C y_{ijc} \log \hat{y}_{ijc},$$

03. Methods

2. Target Image Process

2.



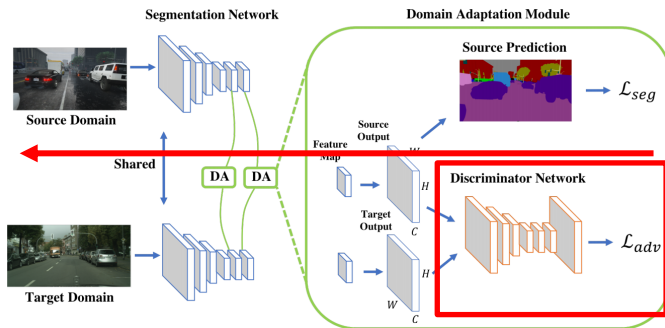
$$\min_{\theta} \frac{1}{N_S} \sum_{k=1}^{N_S} \mathcal{L}_k^S + \frac{1}{N_T} \sum_{k=1}^{N_T} \lambda^T \mathcal{L}_k^T.$$

▶ 기존 UDA Method들의 Loss

03. Methods

2. Target Image Process

► Adversarial Loss



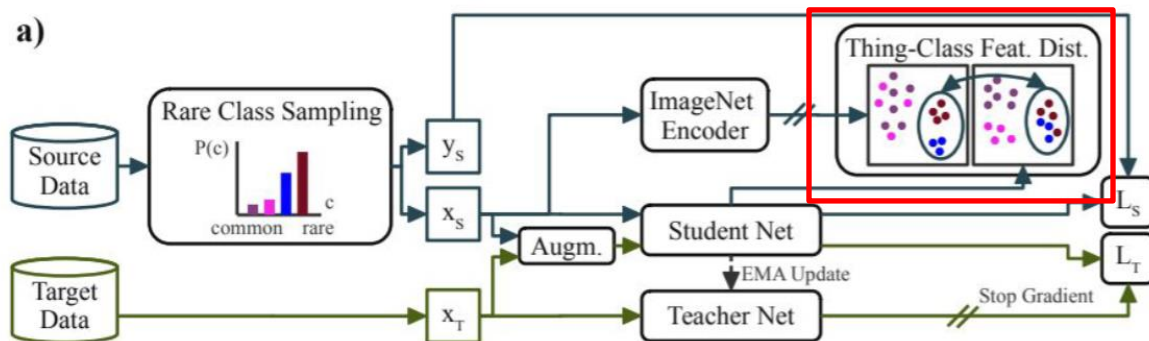
Second, for images in the target domain, we forward them to $\mathbf{G}$ and obtain the prediction $P_t = \mathbf{G}(I_t)$. To make the distribution of P_t closer to P_s , we use an adversarial loss $\mathcal{L}_{adv}$ in (1) as:

$$\mathcal{L}_{adv}(I_t) = - \sum_{h,w} \log(\mathbf{D}(P_t)^{(h,w,1)}). \quad (4)$$

03. Methods

2. Target Image Process

► DAFormer / HRDA



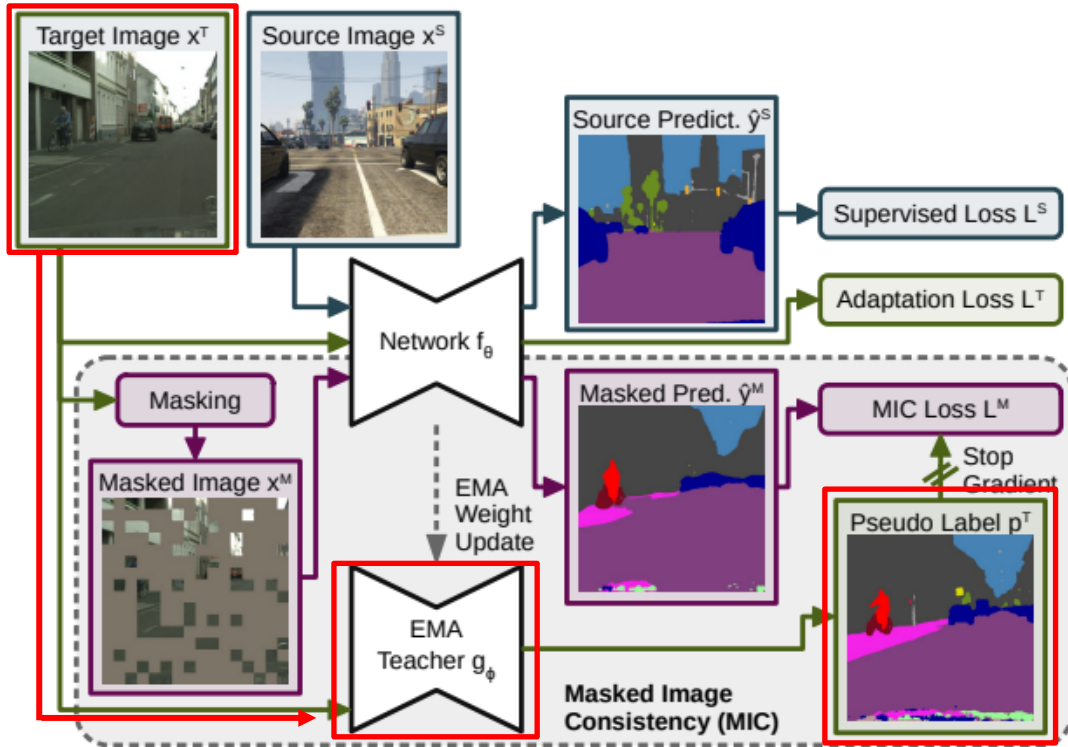
$$d^{(i,j)} = \|F_{ImageNet}(x_S^{(i)})^{(j)} - F_{\theta}(x_S^{(i)})^{(j)}\|_2.$$

$$\mathcal{L}_{FD}^{(i)} = \frac{\sum_{j=1}^{H_F \times W_F} d^{(i,j)} \cdot M_{things}^{(i,j)}}{\sum_j M_{things}^{(i,j)}}$$

03. Methods

3. Target Image Process (MIC)

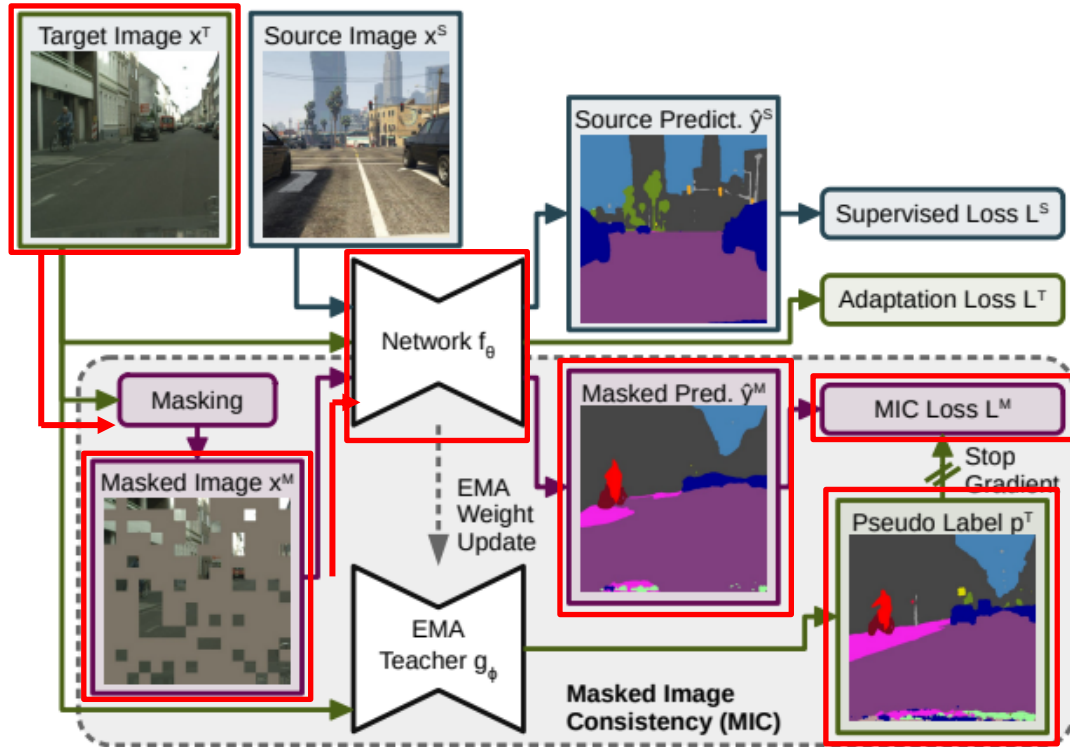
3.



03. Methods

3. Target Image Process (MIC)

4.



$$q^{T,cls} = \max_{c'} g_\phi(x^T)_{c'}$$

$$q^{T,seg} = \frac{\sum_{i=1}^H \sum_{j=1}^W [\max_{c'} g_\phi(x^T)_{(ijc')} > \tau]}{H \cdot W}$$

Quality Weight

$$\mathcal{L}^M = q^T \mathcal{H}(\hat{y}^M, p^T)$$

$$\mathcal{H}(\hat{y}, y) = - \sum_{i=1}^H \sum_{j=1}^W \sum_{c=1}^C y_{ijc} \log \hat{y}_{ijc}$$

03. Methods

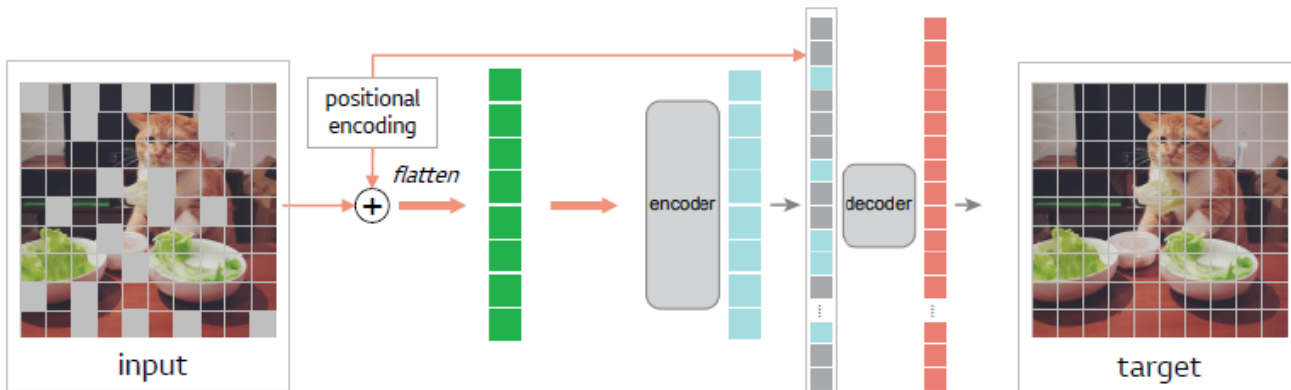
3. Target Image Process (MIC)

▶ MAE (Masked AutoEncoder) Effect

NLP의 BERT : Masking된 부분을 Reconstruction하는 SSL 방법론 → 효과



당장 이미지에 적용해보자



04

Experiments

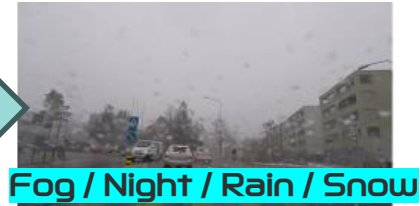
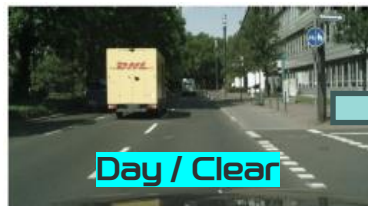
04. Experiments

1. Tasks

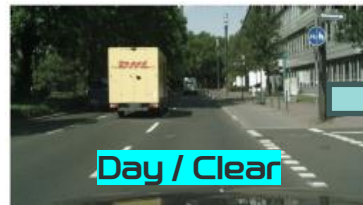
► GTA -> CS (Cityscape)



► CS -> ACDC



► CS -> DarkZurich



04. Experiments

2. Semantic Segmentation

Table 1. Segmentation performance (mIoU in %) of MIC with different UDA methods on GTA→CS.

Network	UDA Method	w/o MIC	w/ MIC	Diff.
DeepLabV2 [5]	Adversarial [84]	44.2	48.2	+4.0
DeepLabV2 [5]	Entropy Min. [87]	44.3	49.0	+4.7
DeepLabV2 [5]	DACS [83]	53.9	56.0	+2.1
DeepLabV2 [5]	DAFormer [32]	56.0	59.4	+3.4
DeepLabV2 [5]	HRDA [33]	63.0	64.2	+1.2
DAFormer [32]	DAFormer [32]	68.3	70.6	+2.3
DAFormer [32]	HRDA [33]	73.8	75.9	+2.1

04. Experiments

2. Semantic Segmentation

Table 2. Semantic segmentation performance (IoU in %) on four different UDA benchmarks.

Method	Road	S.walk	Build.	Wall	Fence	Pole	Tr.Light	Sign	Veget.	Terrain	Sky	Person	Rider	Car	Truck	Bus	Train	M.bike	Bike	mIoU
Synthetic-to-Real: GTA→Cityscapes (Val.)																				
ADVENT [87]	89.4	33.1	81.0	26.6	26.8	27.2	33.5	24.7	83.9	36.7	78.8	58.7	30.5	84.8	38.5	44.5	1.7	31.6	32.4	45.5
DACS [83]	89.9	39.7	87.9	30.7	39.5	38.5	46.4	52.8	88.0	44.0	88.8	67.2	35.8	84.5	45.7	50.2	0.0	27.3	34.0	52.1
ProDA [102]	87.8	56.0	79.7	46.3	44.8	45.6	53.5	53.5	88.6	45.2	82.1	70.7	39.2	88.8	45.5	59.4	1.0	48.9	56.4	57.5
DAFormer [32]	95.7																	9	61.8	68.3
HRDA [33]	96.4																	9	67.5	73.8
MIC (HRDA)	97.4																	5	68.5	75.9
Table 9. Comparison of UDA on GTA→CS and supervised training on CS. “Rel.” indicates $mIoU_{UDA}/mIoU_{Superv.}$																				
ADVENT [87]	85.6																	2	33.0	41.2
DACS [83]	80.6																	5	47.6	48.3
ProDA [102]	87.8																	5	45.6	55.5
DAFormer [32]	84.5																	9	61.7	60.9
HRDA [33]	85.2																	9	64.9	65.8
MIC (HRDA)	86.6																	1	64.3	67.3
ADVENT [87]	85.8																	5	20.7	29.7
MGCDA† [76]	80.3																	1	22.7	42.5
DANNet† [92]	90.0	54.0	74.8	41.0	21.1	25.0	26.8	30.2	72.0	26.2	84.0	47.0	33.9	68.2	19.0	0.3	66.4	38.3	23.6	44.3
DAFormer [32]	93.5	65.5	73.3	39.4	19.2	53.3	44.1	44.0	59.5	34.5	66.6	53.4	52.7	82.1	52.7	9.5	89.3	50.5	38.5	53.8
HRDA [33]	90.4	56.3	72.0	39.5	19.5	57.8	52.7	43.1	59.3	29.1	70.5	60.0	58.6	84.0	75.5	11.2	90.5	51.6	40.9	55.9
MIC (HRDA)	94.8	75.0	84.0	55.1	28.4	62.0	52.6	59.2	46.8	70.0	65.2	61.7	82.1	64.2	18.5	91.3	52.6	44.0	60.2	
Clear-to-Adverse-Weather: Cityscapes→ACDC (Test)																				
ADVENT [87]	72.9	14.3	40.5	16.6	21.2	9.3	17.4	21.2	63.8	23.8	18.3	32.6	19.5	69.5	36.2	34.5	46.2	26.9	36.1	32.7
MGCDA† [76]	73.4	28.7	69.9	19.3	26.3	36.8	53.0	53.3	75.4	32.0	84.6	51.0	26.1	77.6	43.2	45.9	53.9	32.7	41.5	48.7
DANNet† [92]	84.3	54.2	77.6	38.0	30.0	18.9	41.6	35.2	71.3	39.4	86.6	48.7	29.2	76.2	41.6	43.0	58.6	32.6	43.9	50.0
DAFormer [32]	58.4	51.3	84.0	42.7	35.1	50.7	30.0	57.0	74.8	52.8	51.3	58.3	32.6	82.7	58.3	54.9	82.4	44.1	50.7	55.4
HRDA [33]	88.3	57.9	88.1	55.2	36.7	56.3	62.9	65.3	74.2	57.7	85.9	68.8	45.7	88.5	76.4	82.4	87.7	52.7	60.4	68.0
MIC (HRDA)	90.8	67.1	89.2	54.5	40.5	57.2	62.0	68.4	76.3	61.8	87.0	71.3	49.4	89.7	75.7	86.8	89.1	56.9	63.0	70.4

† Method uses additional daytime/clear-weather geographically-aligned reference images.

04. Experiments

2. Semantic Segmentation

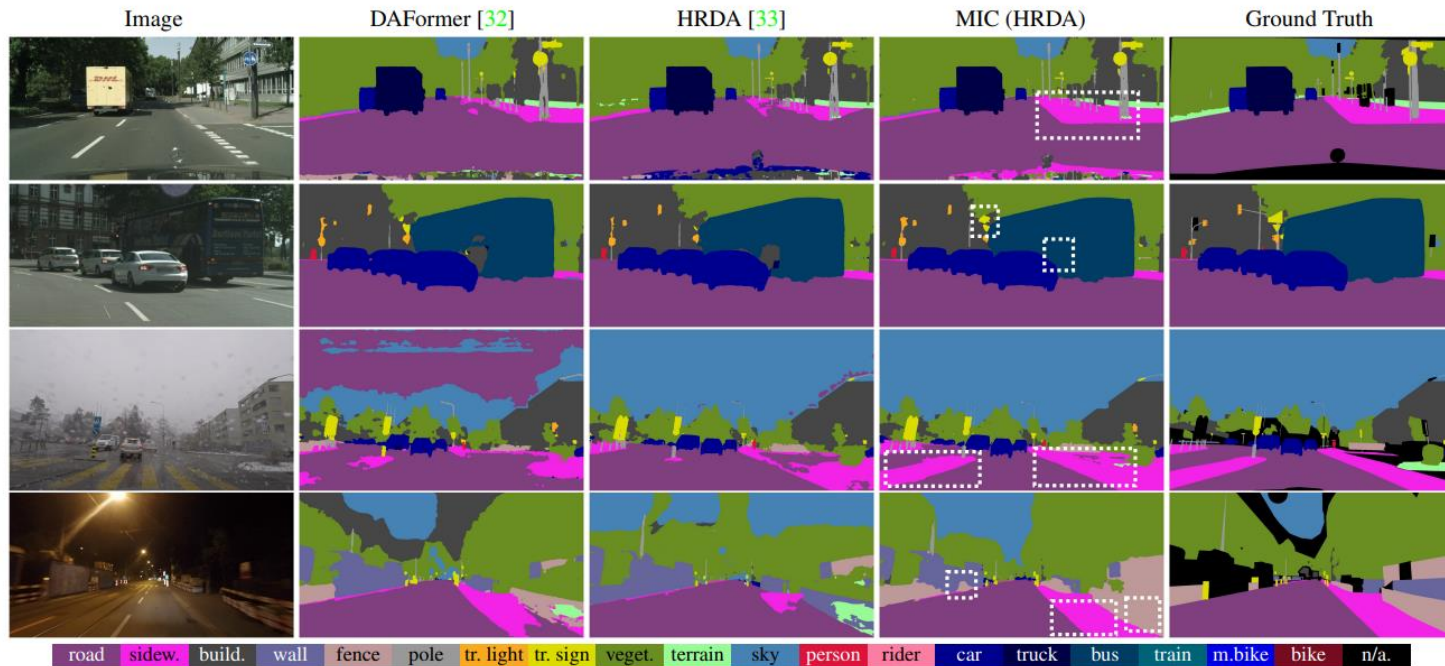


Figure 4. Qualitative comparison of MIC with previous methods on GTA→CS (row 1 and 2), CS→ACDC (row 3), and CS→DarkZurich (row 4). MIC better segments difficult classes such as *sidewalk*, *fence*, *traffic sign*, and *bus*. Further examples are shown in the supplement.

04. Experiments

3. Ablation

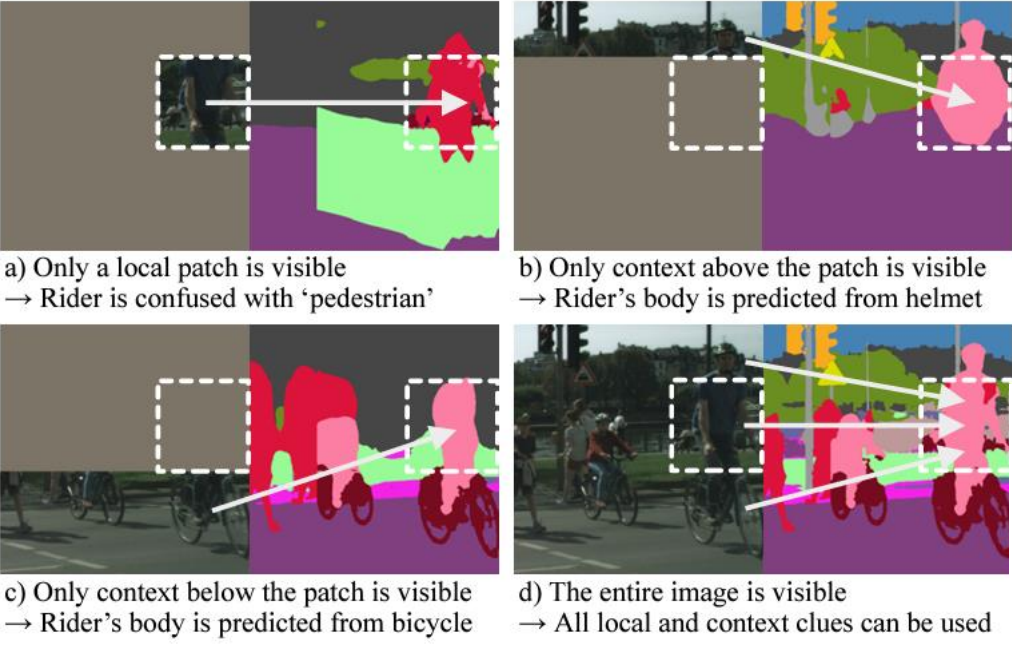


Figure 5. Predictions of MIC for masked variants of the same image demonstrating the learned context priors of MIC.

Patch Size b	Mask Ratio r			
	0.3	0.5	0.7	0.9
32	69.3	69.9	69.7	69.3
64	69.2	70.3	70.6	69.7
128	68.7	70.5	70.4	68.2
256	66.2	69.5	69.8	68.0

Table 7. MIC ablation study with DAFormer [32] on GTA→CS.

	Masked Img.	Color Aug.	EMA Teacher	Pseudo Lbl. Weight	mIoU
1	–	–	–	–	68.3
2	✓	✓	✓	✓	70.6
3	–	✓	✓	✓	50.6
4	✓	–	✓	✓	70.3
5	✓	–	–	✓	69.9
6	✓	✓	✓	–	69.0

05

Conclusion

05. Conclusion

- ▶ UDA의 성능은 Supervised Learning에 근접해짐
- ▶ UDA 기술을 활용해서 지속적으로 알아서 발전하는 형태의 기술이 곧 등장 ?!



-
- ▶ 기존의 Target Domain과 같은 Class / 유사하지만 다른 Distribution의 Domain이 Target
 - ▶ Shape가 많이 다를 경우에 대한 연구도 진행 ?

Q & A

—

Thank you for your attention